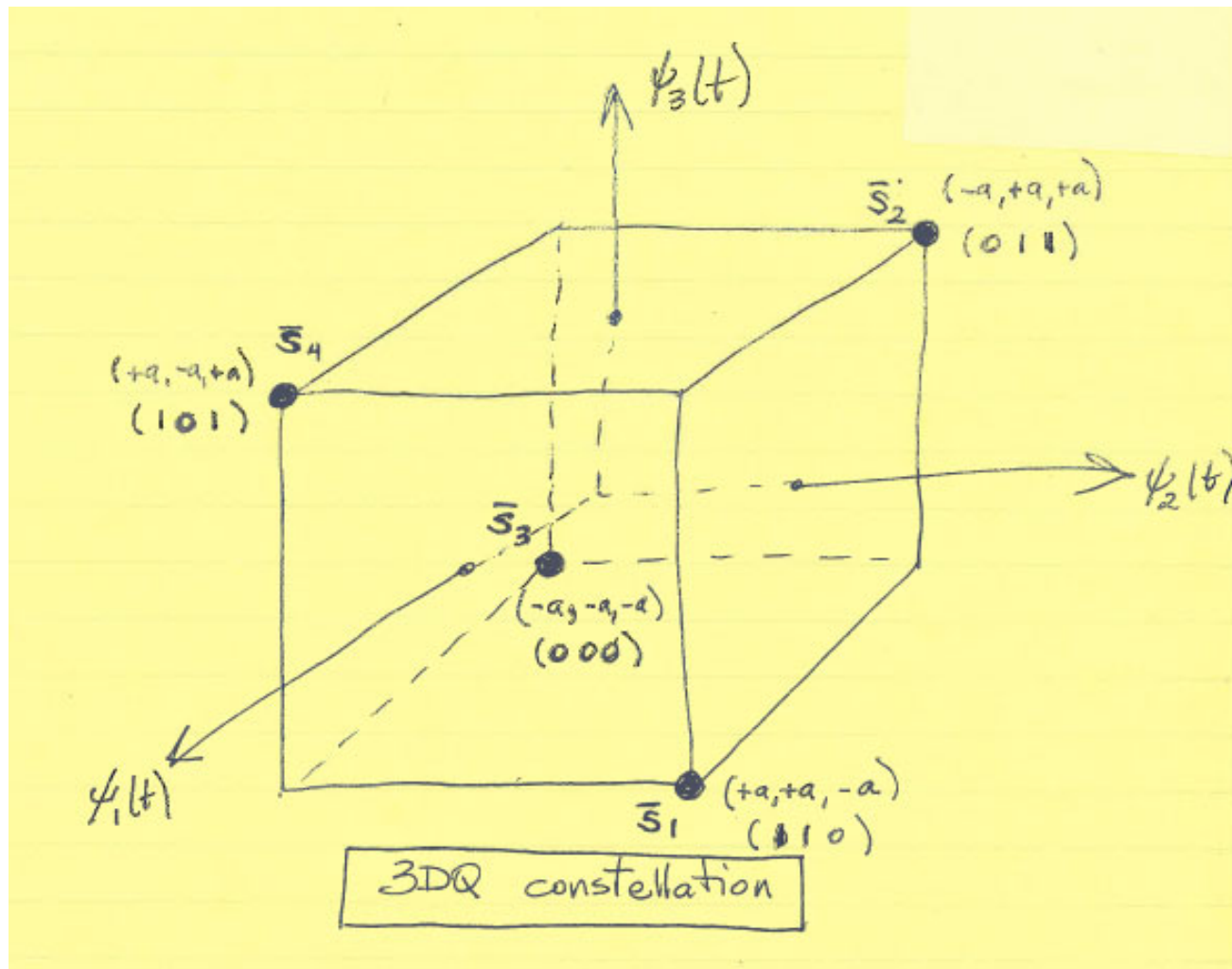
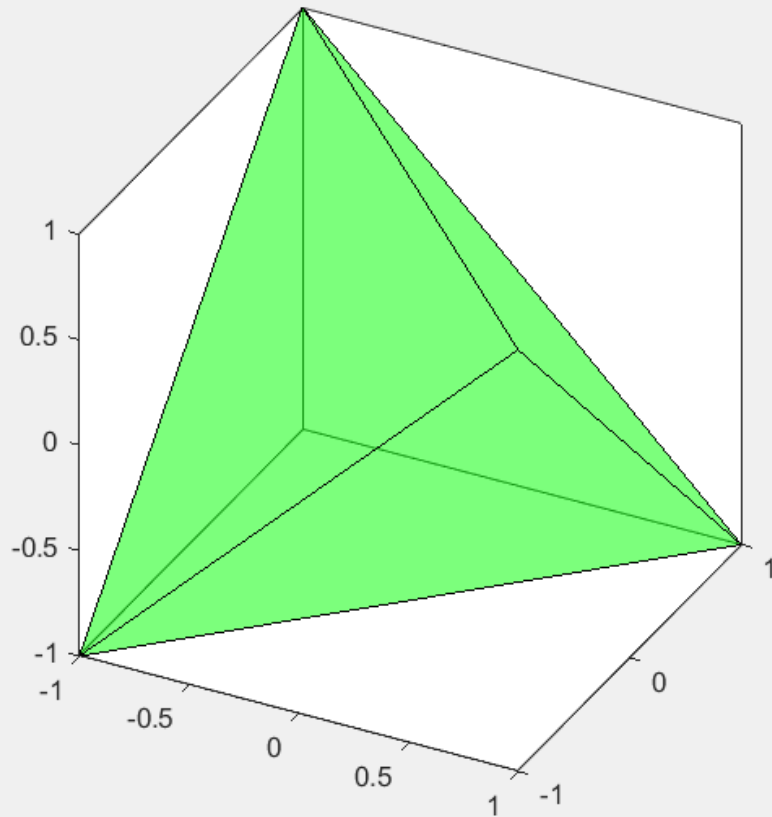
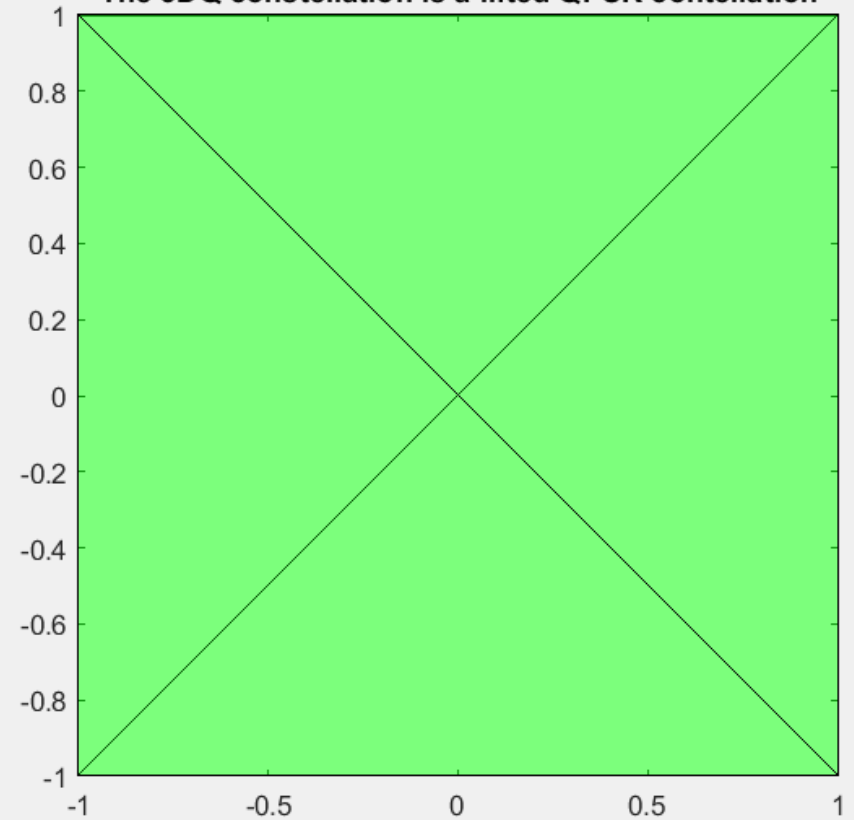




The 3DQ constellation: Vertices of a tetrahedron!



The 3DQ constellation is a lifted QPSK constellation



MATLAB script: "plot_3DQ_constellation.m"

Recall that the syndrome vector, computed from the received vector as $\bar{s} = \bar{r}H^T$ is related to the error vector by the equation

$$\bar{s} = \bar{e}H^T, \quad (1)$$

where H is the parity-check matrix. Note that syndrome $\bar{s}$ is an $(n-k)$ -bit vector.

Eq. (1) above indicates that the syndrome is a function of the error vector, i.e., there is a one-to-one relation between the error vector $\bar{e}$ and the syndrome $\bar{s}$.

The Hamming bound

$$\bar{s} = \bar{e}H^T$$

“To correct up to t errors, the number of different syndrome values $\bar{s}$ needs to be greater than or equal to the number of different error vectors $\bar{e}$ with up to t ones.”

This gives the *Hamming bound*:

$$2^{n-k} \geq \sum_{i=0}^t \binom{n}{i}$$

For $t = 1$, or *single-error correcting codes*, the minimum number of different syndrome values is obtained as

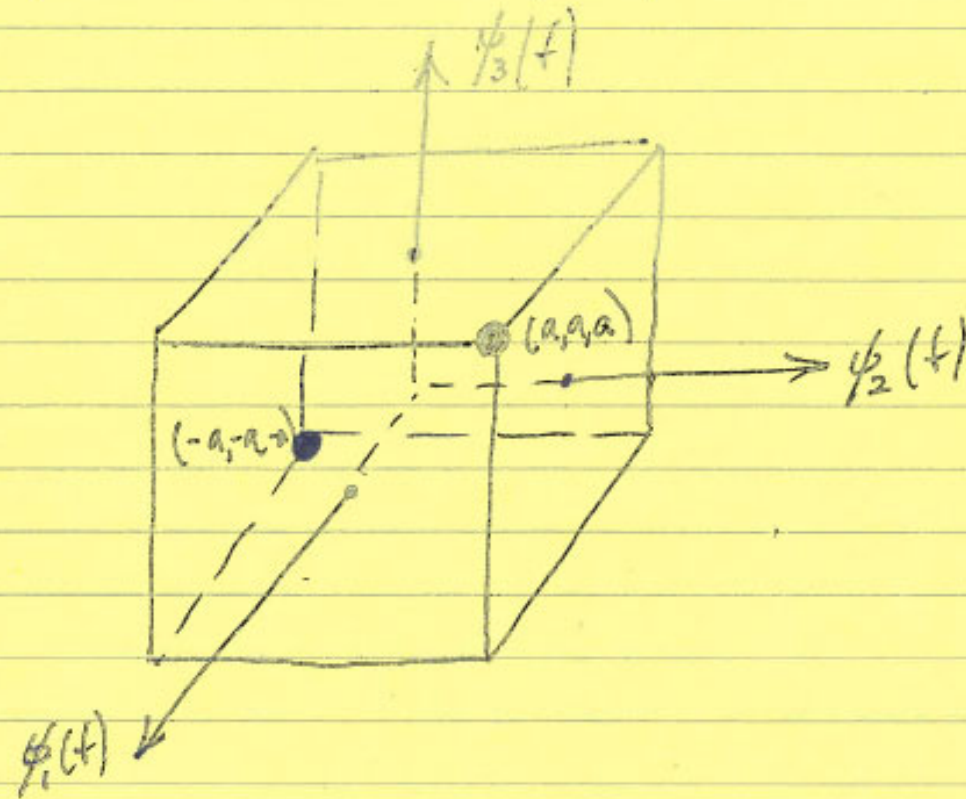
$$2^{n-k} = \sum_{i=0}^1 \binom{n}{i} = 1 + n \quad (2)$$

For a single-bit error correcting code, an equivalent statement is that **all the columns of H be different**. This idea due to Hamming (1949) gives binary linear *Hamming codes*. With $m=n-k$ equal to the number of redundant bits, from Eq. (2) *Hamming codes* have parameters:

Length:	$n = 2^m - 1$
Dimension:	$k = 2^m - 1 - m$
Minimum Hamming distance:	$d_{\min} = 3$

The smallest Hamming code is obtained with $m=2$ and gives the binary linear (3,1,3) repetition code!

Another example: $(3,1,3)$ Binary repetition code



Example: Binary repetition (3,1,3) code

$$G = (111), \quad H = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

<u>Error patterns, $\bar{e}$</u>	<u>Syndrome, $\bar{s} = \bar{e}H^T$</u>
(000)	(00)
(100)	(11)
(010)	(10)
(001)	(01)

Example: If $\bar{C} = (1 1 1)$ and $F = (1 1 0)$
 then $\bar{S} = \bar{F} H^T$

$$= (1 1 0) \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = (0 1)$$

$\longleftrightarrow \bar{e} = (0 0 1)$

From table

$$\therefore \hat{C} = \bar{r} \oplus \bar{e} = \begin{pmatrix} 1 & 1 & 0 \\ \oplus & (0 & 0 & 1) \end{pmatrix} = (1 1 1)$$

Hamming (7,4,3) code:

All nonzero column vectors of 3 bits

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} = [P^T : I_3]$$

Therefore

$$G = [I_4 : P] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Suppose information bits $\bar{B} = (0110)$. Then

$$\bar{C} = (0110) \begin{bmatrix} G \\ + \end{bmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} = \underline{\underline{(0110110)}}$$

Suppose error in 4th position, so that

$$\bar{r} = (0 \ 1 \ 1 \ \underset{\substack{\uparrow \\ \text{error}}}{1} \ 1 \ 1 \ 0)$$

① Compute syndrome:

$$\boxed{\bar{s} = \bar{r} H^T} = (0 \ 1 \ 1 \ 1 \ 1 \ 0) \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{\underline{(1 \ 1 \ 1)}}.$$

Homework:
Write as equations/circuit

② This corresponds to the 4th column of H .
 $\uparrow$ transpose of the
 $\therefore$ Error in 4th position

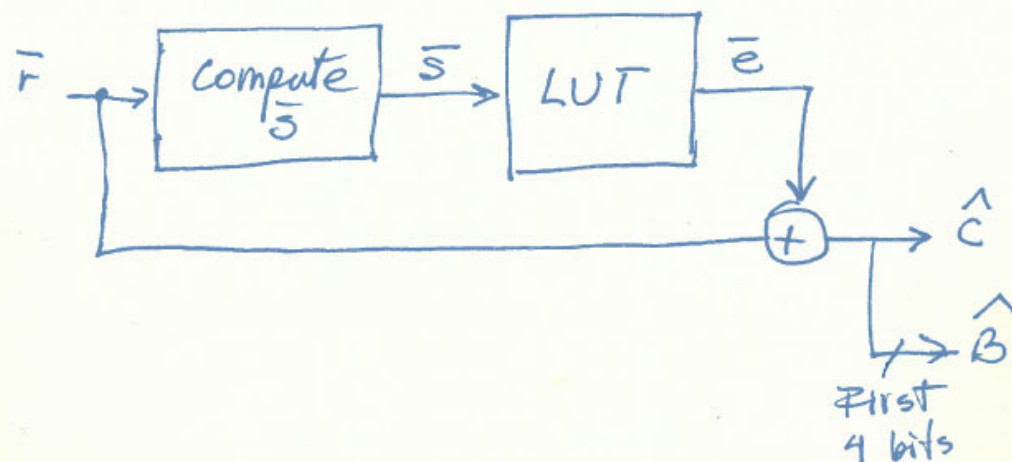
$$\hat{c} = (0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0) \rightarrow (\underbrace{0 \ 1 \ 1}_\text{Information bits} \ 0) = \hat{B}$$

Look-up Table :

Syndrome $\bar{s}$	Error vector $\bar{e}$
000	00000000
110	10000000
011	01000000
101	00100000
111	00010000
100	00001000
010	00000100
001	00000010

H^T points to the Syndrome column.

H-D decoder :



Coded BPSK communication

- Codeword $\bar{c} = (c_1 \ c_2 \ \cdots \ c_n)$ from a binary $(n, k, d_{\min})$ code is transmitted with BPSK modulation (polar mapping) as a vector of **mapper** outputs (signal points) $\bar{s} = (s_1 \ s_2 \ \cdots \ s_n)$, where $s_i = (2c_i - 1)a$, for $i = 1, 2, \dots, n$ and $a = \sqrt{E_s/n}$
- Zero-mean **noise** N_i , with $\sigma_i^2 = N_0/2$, $i = 1, 2, \dots, n$, added at the receiver
- **Matched filter** outputs are $Y_i = s_i + N_i$, $i = 1, 2, \dots, n$
- **Likelihood:**

$$L(\bar{c}) = p(\bar{Y}|\bar{s}) = \frac{1}{(\sqrt{\pi N_0})^n} \exp \left(-\frac{1}{N_0} \sum_{i=1}^n (Y_i - s_i)^2 \right),$$

which is **maximized** by the simplified log likelihood or **correlation**:

$$\ell(\bar{c}) = \sum_{i=1}^n s_i Y_i$$

The log-likelihood (log of the conditional prob.) $\log(p(\bar{Y}|\bar{s})) = L(\bar{c})$ (*)

can be written as

$$L(\bar{c}) = \sum_{i=1}^n s_i Y_i$$

$$\boxed{L(\bar{c}) = \sum_{i=1}^n m(c_i) Y_i}, \quad \underline{\text{correlation}}$$

where Y_i denote MF outputs

Soft-decision decoding

Choose as the most likely transmitted codeword $\bar{c} = \bar{B}G$, the one that maximizes $L(\bar{c})$

$$(*) \quad p(\bar{Y}|\bar{s}) = \frac{1}{(\sqrt{\pi N_0})^n} \cdot e^{-\frac{1}{N_0} \sum_{i=1}^n (Y_i - s_i)^2}$$

$s_i = \pm\sqrt{E}$ for BPSK (polar)

EE161

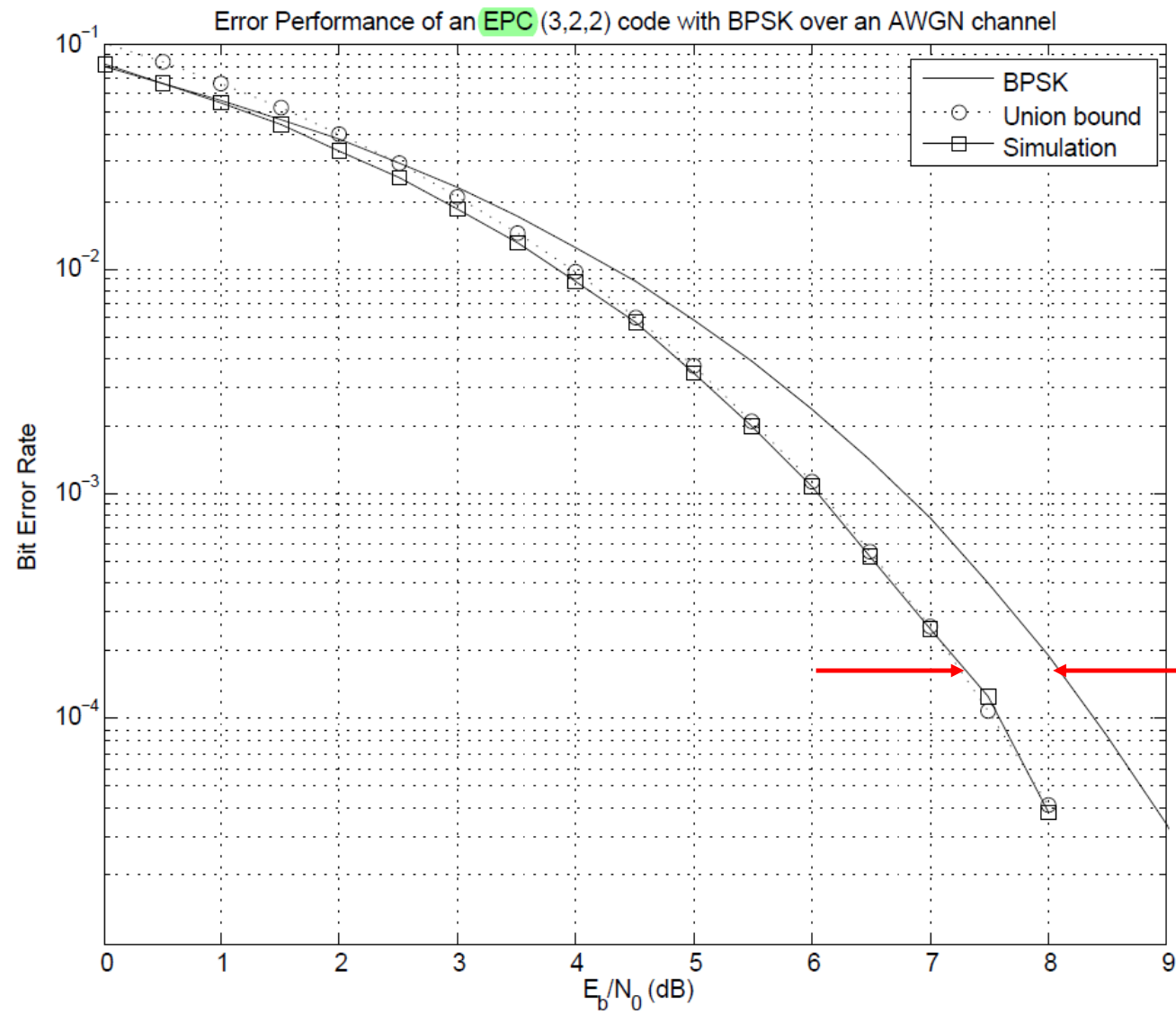
CORRELATIONS OF $(3,2,2)$
EPC CODE

Uncoded $B_1 B_2$	Coded $C_1 C_2 C_3$	$\bar{S}$	Correlations P	Map: $0 \rightarrow +1$ $1 \rightarrow -1$
0 0	0 0 0	a a a	$Y_1 + Y_2 + Y_3$	
0 1	0 1 1	a -a -a	$Y_1 - Y_2 - Y_3$	
1 1	1 1 0	-a -a a	$-Y_1 - Y_2 + Y_3$	
1 0	1 0 1	-a a -a	$-Y_1 + Y_2 - Y_3$	

$$a = \sqrt{E_s/3}$$

MAP: $0 \rightarrow -a$ $1 \rightarrow +a$

$B_1 B_2$	$C_1 C_2 C_3$	$\bar{S}$	Correlations P
0 0	0 0 0	-a -a -a	$-Y_1 - Y_2 - Y_3$
0 1	0 1 1	-a +a +a	$-Y_1 + Y_2 + Y_3$
1 1	1 1 0	+a +a -a	$+Y_1 + Y_2 - Y_3$
1 0	1 0 1	+a -a +a	$+Y_1 - Y_2 + Y_3$



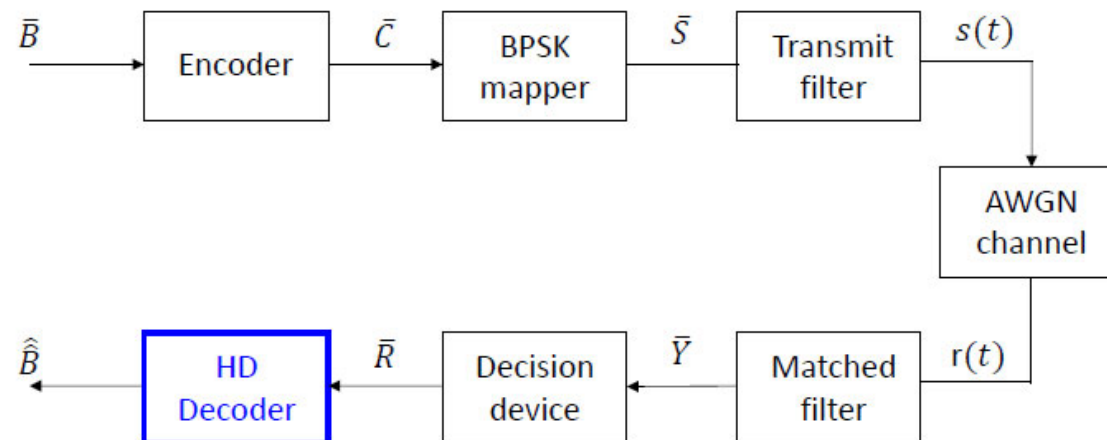
≈ 0.93 dB

Hard-decision (HD) decoding versus soft-decision (SD) decoding

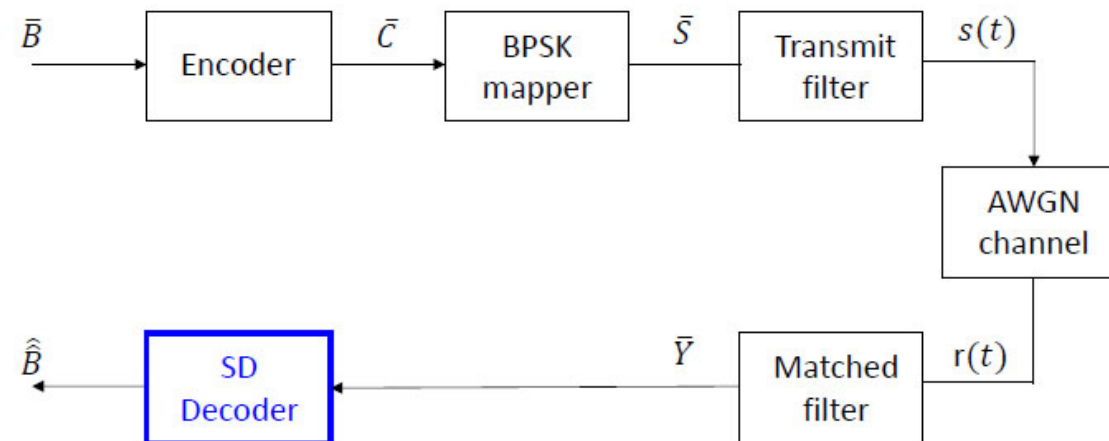
- HD decoding: Correct errors after ML estimation (decision device), R_k
- SD decoding: Use of matched-filter outputs, Y_k

SD decoding performs better than HD decoding, as the *combined likelihood of all coded bits* is used instead of bit-by-bit decisions

Hard-decision decoding

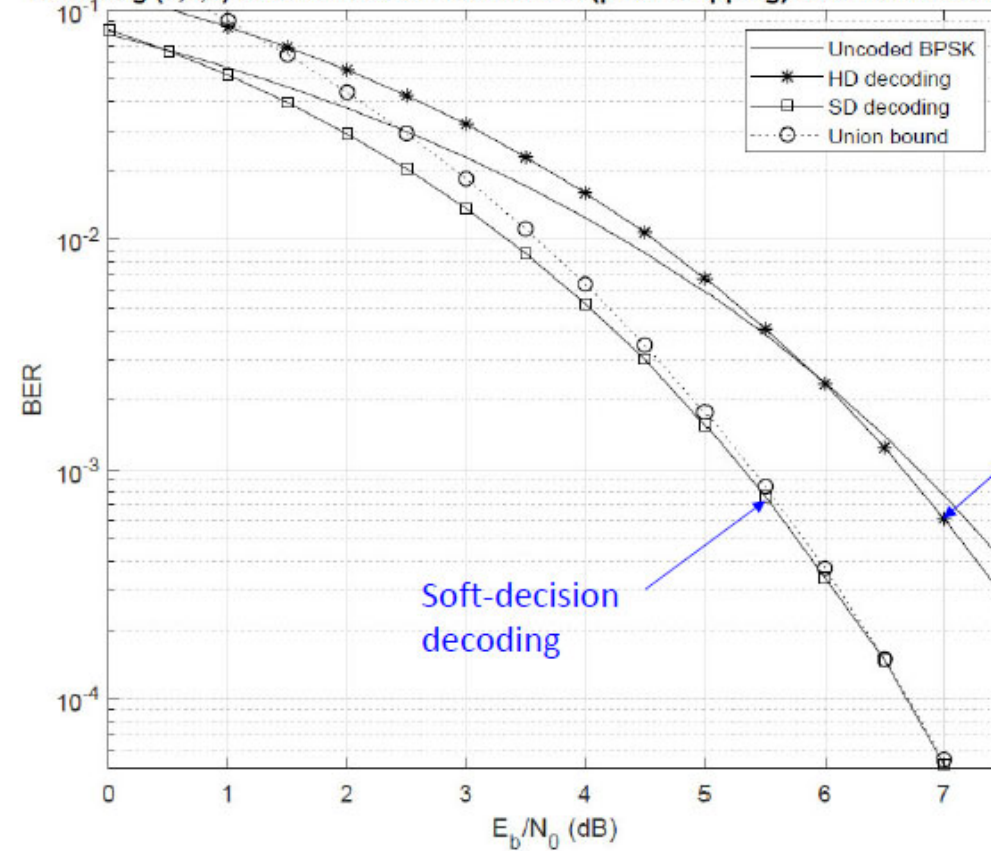


Soft-decision decoding



Example: Binary Hamming (7,4,3) code

Hamming (7,4,3) code with BPSK modulation (polar mapping) over an AWGN channel



Coding gain and "real" (effective) coding gain

Binary $(n, k, d_{\min})$ codes with polar mapping over AWGN

Modulated codeword : $\bar{s} = (\pm a, \pm a, \dots, \pm a)$

$$= a \cdot ((-1)^{b_1}, (-1)^{b_2}, \dots, (-1)^{b_n})$$

Average symbol energy : $n \cdot a^2 = E_s \Rightarrow a = \sqrt{E_s/n}$

Each codeword carries k bits. therefore $E_s = k E_b$
or

$$a = \sqrt{\frac{k}{n} E_b}$$

note, this is
the pulse energy

The ^{min. Euclidean} distance between k codewords modulated is $d_{\min}^E = \sqrt{d_{\min}^2 \cdot a^2}$
 $\sqrt{d_{\min}^2 (2a)^2}$

Finally, the pairwise error probability between modulated codewords is

$$P_2 = Q \left(\sqrt{d_{\min} \frac{k}{n} \frac{2E_b}{N_0}} \right) = Q \left(\sqrt{\frac{d^2}{2N_0}} \right)$$

At high values of E_b/N_0 , the average probability of a bit error

$$P_b \approx Q \left(\sqrt{d_{\min} \cdot \frac{k}{n} \cdot \frac{2E_b}{N_0}} \right) = Q \left(\sqrt{\frac{R \cdot d}{n \cdot N_0}} \right)$$

ASYMPTOTIC CODING GAIN = CG

Code	$d_{\min}$	k/n	CG (dB)
(3, 2, 2)	2	2/3	1.25
(4, 3, 2)	2	3/4	1.76
(7, 4, 3)	3	4/7	2.34

This asymptotic coding gain does not take into account the number of codewords at minimum distance, $A_{\min}$.

The loss due to this number is approx. $0.2 \log_2(A_{\min})$ so that the coding gain can be better as

Real
Coding
Gain

$$RCG \approx 10 \log_{10} \left(d_{\min} \cdot \frac{k}{n} \right) - 0.2 \log_2(A_{\min}) \quad (\text{dB})$$

Code	RCG	$A_{\min}$
$(3,2,2)$	0.93	3
$(4,3,2)$	1.24	6
$(7,4,3)$	1.78	7